

De Bruijn Sequence

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to construct a minimum-length bit string that contains all possible substrings of length n . For example, when $n = 2$, the string 00110 is a valid solution, because its substrings of length 2 are 00, 01, 10 and 11.

Input

The only input line has an integer n .

Output

Print a minimum-length bit string that contains all substrings of length n . You can print any valid solution.

Constraints

- $1 \leq n \leq 15$

Example

Input	Output
2	00110